

Protocol: Rational Design & Directed Evolution Pipeline for Polypropylene Degradation

Version: 5.3

Status: Active

Target: Polypropylene (PP) Oxidation via Engineered GST/Peroxidase Scaffolds

Compute Architecture: CUDA-Accelerated GROMACS (Google Colab Environment)

1. Overview

This protocol defines the standard operating procedure (SOP) for the *in silico* evaluation of enzyme variants targeting the degradation of oxidized polypropylene oligomers. The pipeline integrates structural inference (AlphaFold/Protenix), topology regularization (RDKit/OpenBabel), system solvation (CHARMM-GUI), and high-performance molecular dynamics (GROMACS).

Key Metrics:

- **Stability:** Root Mean Square Deviation (RMSD) of the backbone.
- **Affinity:** Non-bonded Interaction Energy ($E_{int} = E_{Coulomb} + E_{LJ}$).
- **Catalytic Feasibility:** Geometric proximity of substrate to nucleophilic residues (NAC < 3.5 Å).

2. Prerequisites & Dependencies

Computational Resources

- **Runtime Environment:** Google Colab Pro/Pro+ (NVIDIA T4/A100 GPU required).
- **Binary Engine:** MY_GROMACS_GPU.zip (Pre-compiled GROMACS 2023.x with CUDA support).
- **Scripts:**
 1. 01_Topology_Preprocessor.ipynb
 2. 02_HPC_Simulation.ipynb
 3. 03_PostHoc_Analysis.ipynb

Chemical Definitions

- **Substrate:** Oxidized Polypropylene Trimer (4-methylheptan-2-one derivative).

- **SMILES:** CC(C)CC(=O)CC(C)C
- **Cofactor:** L-Glutathione (Reduced, for GST coupling).
 - **SMILES:** C(CC(=O)N[C@@H](CS)C(=O)NCC(=O)O)[C@@H](C(=O)O)N

3. Workflow Execution

Phase I: Structural Inference (Upstream)

Objective: Generation of ternary complex coordinates (Enzyme + Cofactor + Substrate).

1. **Platform:** Protenix / AlphaFold 3 Server.
2. **Input Configuration:**
 - **Sequence:** Target Variant FASTA.
 - **Ligands:** Explicit definition of PP and GSH.
3. **Deliverable:** .cif structure file (e.g., wt_GST01_PP_v3_sample_0.cif).
 - *Note:* Ensure filename consistency with the TARGET_MAP variable in Phase II.

Phase II: Topology Regularization

Objective: Stereochemical correction and forcefield topology generation. **Script:**

01_Topology_Preprocessor.ipynb

1. **Input:** Upload the Phase I .cif file.
2. **Module Execution:**
 - **Ligand Conformational Sampling:** Generates 50 conformers per ligand using ETKDGV3; minimizes energy via MMFF94.
 - **Topology Alignment:** Aligns optimized ligands to the target crystal structure coordinates.
 - **Standardization:** Renames atoms (C1, C2...) to ensure CHARMM forcefield compatibility.
3. **Artifacts (Download Required):**
 - final_system_v5_3.pdb (Master Coordinate File)
 - plastic_final3.mol2 (Substrate Topology)
 - gsh_final3.mol2 (Cofactor Topology)

Phase III: System Construction (CHARMM-GUI)

Objective: Solvation, ionization, and parameter assignment. **Platform:** [CHARMM-GUI Solution Builder](#)

1. Upload Configuration:

- Input: `final_system_v5_3.pdb` .
- **Hetero Chain Handling:** Select "Upload RTF/MOL2".
 - Map LIG → `plastic_final3.mol2` .
 - Map GSH → `gsh_final3.mol2` .

2. Parameterization:

- Forcefield: **CHARMM36m** (Standard Protein/Lipid/Ligand).
- Ligand Reader: CGenFF (Verify penalty scores < 50).

3. Simulation Box:

- Geometry: Rectangular.
- Edge Distance: 10 Å (Minimum Image Convention buffer).
- Fit: "Fit to Protein".

4. Solvent Environment:

- Ion Placement: Monte-Carlo.
- Ionic Strength: 0.15 M KCl (Physiological).

5. Output: Download `charmm-gui.tgz` .

Phase IV: Production Dynamics (GPU)

Objective: Thermodynamic equilibration and trajectory generation. **Script:**

`02_HPC_Simulation.ipynb`

1. Environment Initialization:

- Upload `MY_GROMACS_GPU.zip` (Engine).
- Upload `charmm-gui.tgz` (System).

2. Pipeline Execution:

- **Minimization:** Steepest Descent (converge to $F_{max} < 1000$ kJ/mol/nm).
- **Equilibration:** NVT/NPT ensembles with positional restraints.
- **Telemetry:** Monitor ns/day to verify GPU offloading.

3. Production Run:

- Ensemble: NPT (303.15 K, 1 bar).
- Duration: 1 ns (Benchmarking) / 100 ns (Production).
- *Auto-Process:* `trjconv` removes solvent and centers the protein.

4. Data Archival:

- Script generates `FINAL_SIMULATION_DATA.zip`.
- **Action:** Download immediately.

Phase V: Quantitative Auditing

Objective: Geometric and energetic validation of the complex. **Script:**

`03_PostHoc_Analysis.ipynb`

1. **Input:** Upload `FINAL_SIMULATION_DATA.zip`.

2. System Audit:

- Parses `.gro` to classify residues.
- *Pass Criteria:* Detection of `LIG` residues with Carbon atoms.

3. Interaction Energy (E_{int}):

- Recalculates short-range non-bonded energies (LJ + Coulomb).
- **Thresholds:**
 - $|E| < 5.0$ kJ/mol $\rightarrow$ **Dissociation (Fail).**
 - $|E| > 50.0$ kJ/mol $\rightarrow$ **Stable Complex (Pass).**

4. Catalytic Proximity (NAC Analysis):

- Measures minimal distance (d_{min}) between Substrate and Nucleophile (Cys/Ser/His/Tyr).
- **Classifications:**
 - **Reactive (NAC):** $d_{min} < 3.5$ Å.
 - **Metastable:** 3.5 Å $< d_{min} < 5.0$ Å.
 - **Non-Reactive:** $d_{min} > 5.0$ Å.

4. Software Architecture & References

The validity of this *in silico* pipeline relies on the following open-source algorithms and web servers.

Core Engines

- **GROMACS (2023.x):** High-performance molecular dynamics engine.
 - *Citation:* Abraham, M. J., et al. (2015). *SoftwareX*, 1-2, 19-25.
 - *URL:* <https://www.gromacs.org/>
- **AlphaFold 3 / Protenix:** Deep learning-based protein structure prediction.

- *Citation:* Jumper, J., et al. (2021). *Nature*, 596, 583–589.

Topology & Cheminformatics

- **CHARMM-GUI:** Automated simulation system builder and forcefield assignment.
 - *Citation:* Jo, S., et al. (2008). *J. Comput. Chem.*, 29(11), 1859–1865.
 - *URL:* <https://www.charmm-gui.org/>
- **RDKit:** Open-source cheminformatics for conformer generation and SMILES parsing.
 - *URL:* <https://www.rdkit.org/>
- **Open Babel:** Chemical toolbox for file format interconversion.
 - *Citation:* O'Boyle, N. M., et al. (2011). *J. Cheminform.*, 3, 33.

Analysis Libraries

- **MDTraj:** Trajectory analysis and geometric calculations.
 - *Citation:* McGibbon, R. T., et al. (2015). *Biophys. J.*, 109(8), 1528–1532.
- **CGenFF:** CHARMM General Force Field for organic ligands.